

CC Lab 13

Task 1:

```
1: Start: a = 0
2: b = 1
3: if a > b goto L1
4: c = a + b
5: goto L2
6: L1: c = a - b
7: L2: d = c * 2
8: print d
```

(a) Leaders:

Leaders are:

1 → first instruction

3 → contains jump (conditional)

6 (L1) → target of jump

7 (L2) → target of jump

Final leaders: 1, 3, 6, 7

(b) Basic Blocks:

B1: [1, 2]

B2: [3, 4, 5]

B3: [6]

B4: [7, 8]

GIVENCHY

(c) Flow Graph (Connections)

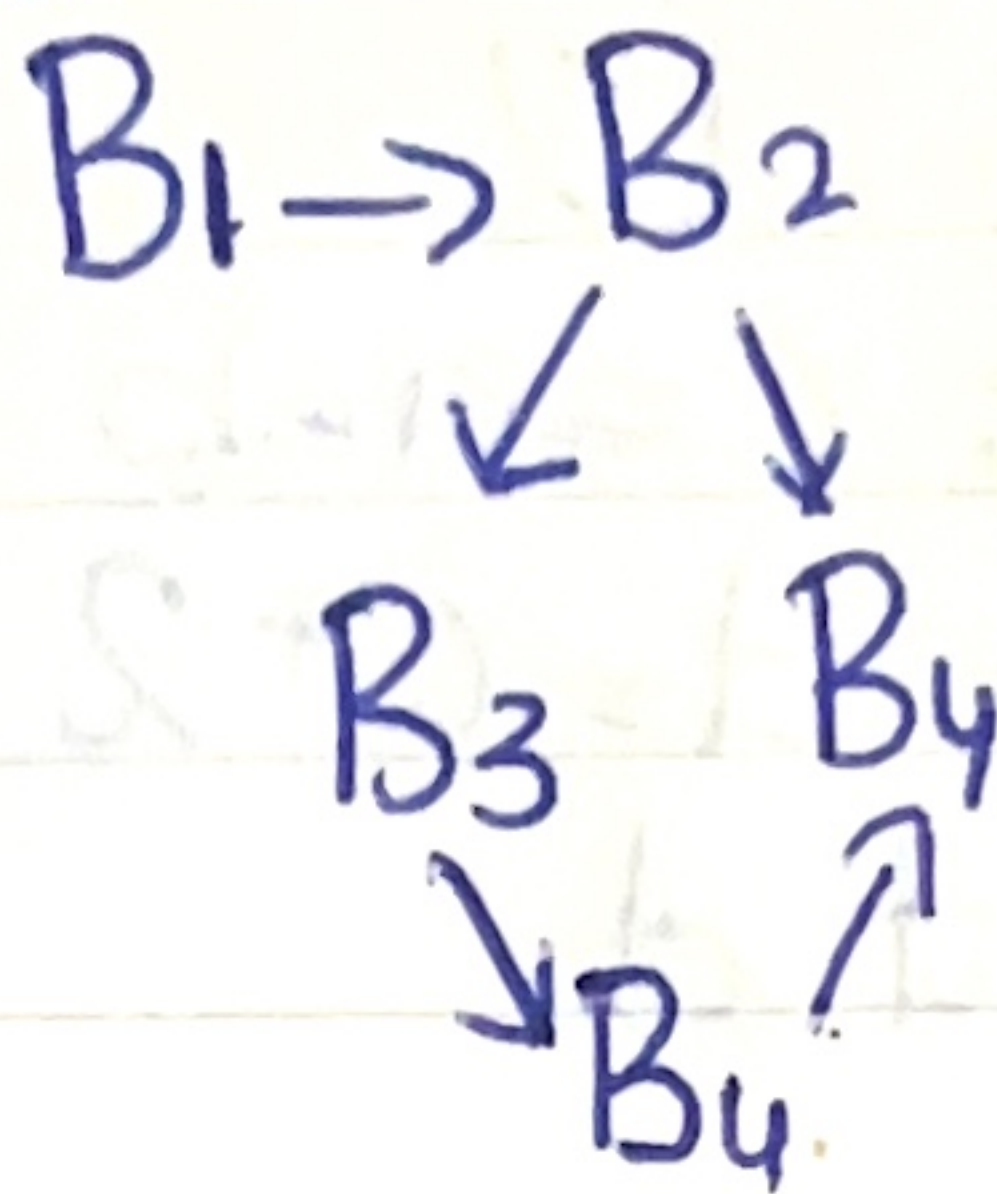
$B_1 \rightarrow B_2$

$B_2 \rightarrow B_3$ (if condition TRUE)

$B_2 \rightarrow B_4$ (if condition FALSE via goto L2)

$B_3 \rightarrow B_4$

~~Structure:~~ Structure:



Task 2:

$$t_1 = 5 * 2$$

$$t_2 = t_1 + 3$$

$$a = t_2$$

$$t_3 = a * 2$$

$$b = t_3$$

$$c = a + 2$$

Step-by-Step Optimization:

1. Constant Folding:

$$t_1 = 10$$

$$t_2 = 13$$

2. Propagation:

$$a = 13$$

$$t_3 = 26$$

$$b = 26$$

$$c = 15$$

3. Dead Code Elimination (remove unused temp)

Final Optimized Code:

$$a = 13$$

$$b = 26$$

$$c = 15$$

Task 3:

$$i = 0$$

L: $t_1 = 10 * 20$ // constant

$$t_2 = t_1 + i$$

$t_3 = p * q$ // p & q never change inside loop

$$a[i] = t_2 + t_3$$

$$i = i + 1$$

if $i < 100$ goto L

After Optimization:

$$t1 = 200$$

$$t3 = p * a$$

$$i = 0$$

$$L: t2 = t1 + i$$

$$a[i] = t2 + t3$$

$$i = i + 1$$

if $i < 100$ goto L

Task 4:

Statement

1. $x = y * 1 \rightarrow x = y$

2. Move $c = a * b$ outside loop

3. $4 * i \rightarrow t = t + 4$

4. $t1 = 8 + 5 \rightarrow 13$

5. Remove unused $x = 3$

Optimization Type

Algebraic Simplification

Loop-Invariant Code Motion

Strength Reduction

Constant Folding

Dead code elimination.